



Strathmore
UNIVERSITY

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STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS
CAT 2 - MARKING SCHEME
DSA 8505: Bayesian Statistics

DATE: 17 Feb 2026

Instruction

- (a) Answer All Question
- (b) Scan and submit your answer sheet through the e-learning by 21h59, 17th Feb 2026.

1. A security agency uses an automated facial recognition system to identify individuals on a watchlist at a busy airport. Based on historical records, it is known that only 0.5% of travelers are actually on the watchlist.

The facial recognition system has the following performance characteristics:

- If a traveler has the disease, the system correctly flags them with probability 95%.
- If a traveler does not have the disease, the system falsely flags them with probability 3%.

A randomly selected traveler is flagged as a match.

- (a) Define the prior probability that a randomly selected traveler is on the watchlist.

Solution

The prior probability represents the probability that a traveler is on the watchlist *before* observing the facial recognition result.

Since historical records indicate that only 0.5% of travelers are on the watchlist, the prior probability is

$$P(W) = 0.005,$$

where W denotes the event that a traveler is on the watchlist.

Consequently, the probability that a traveler is *not* on the watchlist is

$$P(W^c) = 1 - 0.005 = 0.995.$$

✓ ①

- (b) Define the likelihood of the system flagging a traveler, conditional on whether the traveler is on the watchlist or not.

Solution

The likelihood specifies the probability of observing the test result given the true status of the traveler.

Let F denote the event that the system flags a traveler.

- If the traveler is on the watchlist, the system flags them with probability

$$P(F | W) = 0.95. \quad \checkmark (1)$$

- If the traveler is not on the watchlist, the system incorrectly flags them with probability

$$P(F | W^c) = 0.03. \quad \checkmark (1)$$

These conditional probabilities constitute the likelihood model for the facial recognition system.

- (c) Compute the marginal probability that a randomly selected traveler is flagged by the system.

Solution

The marginal probability of a traveler being flagged is obtained using the law of total probability:

$$P(F) = P(F | W)P(W) + P(F | W^c)P(W^c). \quad \checkmark (1)$$

Substituting the given values:

$$P(F) = (0.95)(0.005) + (0.03)(0.995) \quad \checkmark (1)$$

$$P(F) = 0.00475 + 0.02985 = 0.0346.$$

Thus, the probability that a randomly selected traveler is flagged by the system is

$$P(F) = 0.0346. \quad \checkmark (1)$$

- (d) Using Bayes' Theorem, compute the posterior probability that the traveler is actually on the watchlist given that they were flagged by the system.

Solution

Bayes' Theorem gives the posterior probability as

$$P(W | F) = \frac{P(F | W)P(W)}{P(F)}. \quad \checkmark (1)$$

Substituting the computed quantities:

$$P(W | F) = \frac{(0.95)(0.005)}{0.0346}. \quad \checkmark (1)$$

$$P(W | F) = \frac{0.00475}{0.0346} \approx 0.137.$$

Therefore, the posterior probability that a flagged traveler is actually on the watchlist is

$$P(W | F) \approx 0.137 \text{ (13.7\%)}$$

✓ ①

(e) Interpret the result from a policy perspective.

Solution

Although the facial recognition system has high sensitivity (95%), the posterior probability that a flagged traveler is truly on the watchlist is only about 13.7%. This relatively low probability is due to the extremely small prior probability (0.5%) that any given traveler is on the watchlist, illustrating the *base-rate effect*.

From a policy perspective, this suggests that immediately detaining all flagged travelers would result in a large number of false positives. A more appropriate approach would be to subject flagged individuals to additional screening or verification procedures before taking further action.

✓ ①

2. Consider the problem of modelling monthly electricity consumption y using average monthly temperature x .

- (a) Using the ordinary least squares (OLS) method, estimate the regression coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ for Model M_1 .

Solution

Model M_1 is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Step 1: Compute sample means

$$\bar{x} = \frac{10 + 15 + 20 + 25 + 30}{5} = 20, \quad \bar{y} = \frac{220 + 240 + 280 + 350 + 460}{5} = 310.$$

Step 2: Compute slope estimator

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}.$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 3000, \quad \sum (x_i - \bar{x})^2 = 250.$$

$$\hat{\beta}_1 = \frac{3000}{250} = 12.$$

Step 3: Compute intercept

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 310 - 12(20) = 70.$$

$$\boxed{\hat{\beta}_0 = 70, \quad \hat{\beta}_1 = 12}$$

- (b) Write down the likelihood function for Model M_1 .

Solution

Under Model M_1 , the relationship between electricity consumption and temperature is assumed to be linear:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where the random errors satisfy

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n.$$

Step 1: Specify the conditional distribution of each observation

Conditioned on the parameters β_0 , β_1 , and σ^2 , each observation y_i follows a Normal distribution with mean $\beta_0 + \beta_1 x_i$ and variance σ^2 :

$$y_i \mid \beta_0, \beta_1, \sigma^2, M_1 \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2).$$

Thus, the conditional density of a single observation is

$$p(y_i \mid \beta_0, \beta_1, \sigma^2, M_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right]. \quad (1)$$

Step 2: Use the independence assumption

The regression model assumes that the observations are conditionally independent given the parameters. Therefore, the joint density of the data vector $\mathbf{y} = (y_1, \dots, y_n)^T$ is the product of the individual densities:

$$p(\mathbf{y} | \beta_0, \beta_1, \sigma^2, M_1) = \prod_{i=1}^n p(y_i | \beta_0, \beta_1, \sigma^2, M_1).$$

Step 3: Write the likelihood function

Substituting the Normal density into the product gives the likelihood function:

$$p(\mathbf{y} | \beta_0, \beta_1, \sigma^2, M_1) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right].$$

Step 4: Likelihood up to proportionality

For Bayesian inference, the likelihood is often expressed up to proportionality with respect to the parameters:

$$p(\mathbf{y} | \beta_0, \beta_1, \sigma^2, M_1) \propto (\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right]. \quad (2)$$

- (c) Using Bayes' theorem, write the posterior distribution of the regression parameters under Model M_2 .

Solution

For Model M_2 , the regression model is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n.$$

Let

$$\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^T, \quad \mathbf{y} = (y_1, \dots, y_n)^T.$$

Step 1: Likelihood distribution

Conditional on $\boldsymbol{\beta}$ and σ^2 , the observations are independent and normally distributed:

$$y_i | \boldsymbol{\beta}, \sigma^2, M_2 \sim \mathcal{N}(\beta_0 + \beta_1 x_i + \beta_2 x_i^2, \sigma^2).$$

Hence, the likelihood function is

$$p(\mathbf{y} | \boldsymbol{\beta}, \sigma^2, M_2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2)^2}{2\sigma^2}\right]. \quad (1)$$

Step 2: Prior distributions

The priors for the regression coefficients are independent Normal distributions:

$$\beta_j \sim \mathcal{N}(0, 10^2), \quad j = 0, 1, 2,$$

which implies

$$p(\boldsymbol{\beta}) = \prod_{j=0}^2 \frac{1}{\sqrt{2\pi(10^2)}} \exp\left(-\frac{\beta_j^2}{2(10^2)}\right).$$

The prior for the error variance is

$$\sigma^2 \sim \text{Inverse-Gamma}(1, 1),$$

with density

$$p(\sigma^2) = \frac{1}{\Gamma(1)} (\sigma^2)^{-2} \exp\left(-\frac{1}{\sigma^2}\right), \quad \sigma^2 > 0.$$

Step 3: Posterior distribution via Bayes' theorem

Bayes' theorem gives

$$p(\beta, \sigma^2 | \mathbf{y}, M_2) \propto p(\mathbf{y} | \beta, \sigma^2, M_2) p(\beta) p(\sigma^2).$$

Substituting the likelihood and priors:

$$p(\beta, \sigma^2 | \mathbf{y}, M_2) \propto (\sigma^2)^{-\frac{n}{2}} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2)^2\right] \\ \times \exp\left(-\frac{1}{2(10^2)} \sum_{j=0}^2 \beta_j^2\right) \times (\sigma^2)^{-2} \exp\left(-\frac{1}{\sigma^2}\right)$$

Step 4: Final posterior form

Combining terms, the joint posterior density is

$$p(\beta, \sigma^2 | \mathbf{y}, M_2) \propto (\sigma^2)^{-\left(\frac{n}{2}+2\right)} \exp\left[-\frac{1}{2\sigma^2} \left\{ \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2)^2 + 2 \right\}\right] \\ \exp\left(-\frac{1}{2(10^2)} \sum_{j=0}^2 \beta_j^2\right).$$

(d) Explain why Markov Chain Monte Carlo (MCMC) methods are required.

Solution

The posterior distribution under Model M_2 involves:

- multiple regression parameters,
- a non-linear design matrix due to the x^2 term,
- integration over σ^2 .

The resulting posterior has no closed-form normalizing constant. Therefore, direct sampling is infeasible and numerical integration is impractical. MCMC methods (such as Gibbs sampling or Metropolis Hastings) are required to generate samples from the posterior distribution and approximate posterior summaries.

(e) Model comparison using marginal likelihoods.

(i) Compute the Bayes Factor in favour of Model M_2 .

Solution

The Bayes Factor provides a quantitative measure of the evidence in the data in favour of one model over another.

Step 1: Definition of the Bayes Factor

For two competing models M_1 and M_2 , the Bayes Factor in favour of Model M_2 against Model M_1 is defined as

$$BF_{21} = \frac{P(D | M_2)}{P(D | M_1)},$$

where $P(D | M_k)$ is the *marginal likelihood* (or *model evidence*) under Model M_k .

Step 2: Substitute the given marginal likelihoods

The marginal likelihoods are provided as

$$P(D | M_1) = 0.015, \quad P(D | M_2) = 0.045.$$

Substituting these values into the Bayes Factor formula:

$$BF_{21} = \frac{0.045}{0.015}.$$

Step 3: Compute the numerical value

$$BF_{21} = 3.$$

Step 4: State the result

$$BF_{21} = 3$$

Interpretation

A Bayes Factor of 3 means that the observed data are three times more likely under Model M_2 than under Model M_1 . According to common interpretative guidelines (e.g., Jeffreys' scale), this corresponds to *moderate evidence* in favour of the quadratic temperature model M_2 .

- (ii) Calculate the posterior model probabilities.

Solution

Using Bayes' theorem:

$$P(M_i | D) = \frac{P(D | M_i)P(M_i)}{P(D | M_1)P(M_1) + P(D | M_2)P(M_2)}.$$

$$P(M_1 | D) = \frac{0.015 \times 0.5}{0.03} = 0.25, \quad P(M_2 | D) = \frac{0.045 \times 0.5}{0.03} = 0.75.$$

- (iii) Interpret the model comparison.

Solution

Model M_2 has three times more support from the data than Model M_1 . The posterior probability of 0.75 indicates strong evidence in favour of the quadratic temperature model, suggesting a non-linear relationship between temperature and electricity consumption.

(f) Suppose the posterior mean estimates of the regression coefficients are obtained as:

- Model M_1 : $\hat{\beta}_0 = 150$, $\hat{\beta}_1 = 8.5$
- Model M_2 : $\hat{\beta}_0 = 120$, $\hat{\beta}_1 = 6.2$, $\hat{\beta}_2 = 0.45$

(i) Write down the general BMA formula.

Solution

The BMA estimator for a regression coefficient β_j is

$$\hat{\beta}_j^{\text{BMA}} = \sum_k E(\beta_j | M_k, D) P(M_k | D),$$

where the sum is over all candidate models.

(ii) Compute the BMA estimates.

Solution

$$\hat{\beta}_0^{\text{BMA}} = 150(0.25) + 120(0.75) = 127.5,$$

$$\hat{\beta}_1^{\text{BMA}} = 8.5(0.25) + 6.2(0.75) = 6.78,$$

$$\hat{\beta}_2^{\text{BMA}} = 0(0.25) + 0.45(0.75) = 0.3375.$$

(iii) Write the final BMA regression equation.

Solution

The Bayesian Model Averaged regression equation is

$$\hat{y} = 127.5 + 6.78x + 0.3375x^2.$$